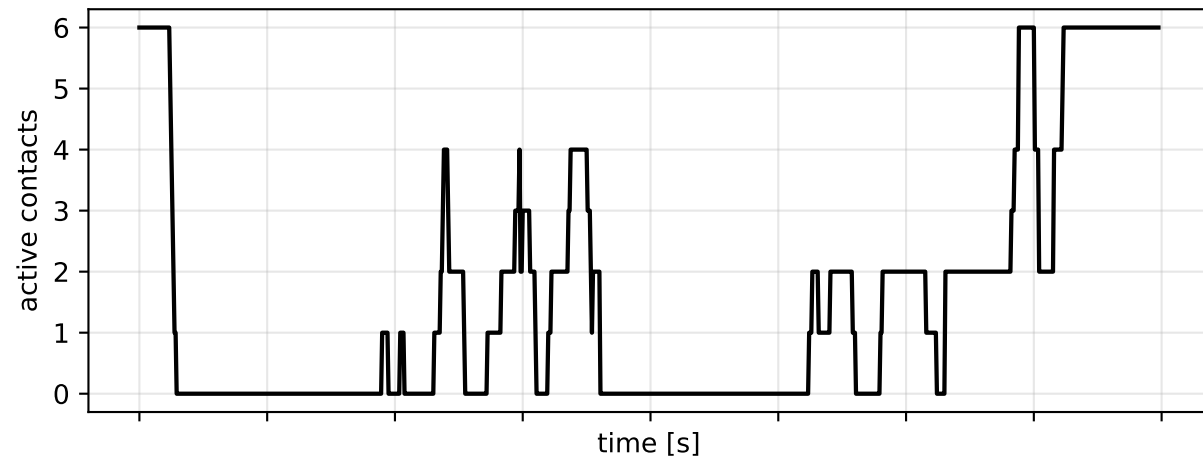
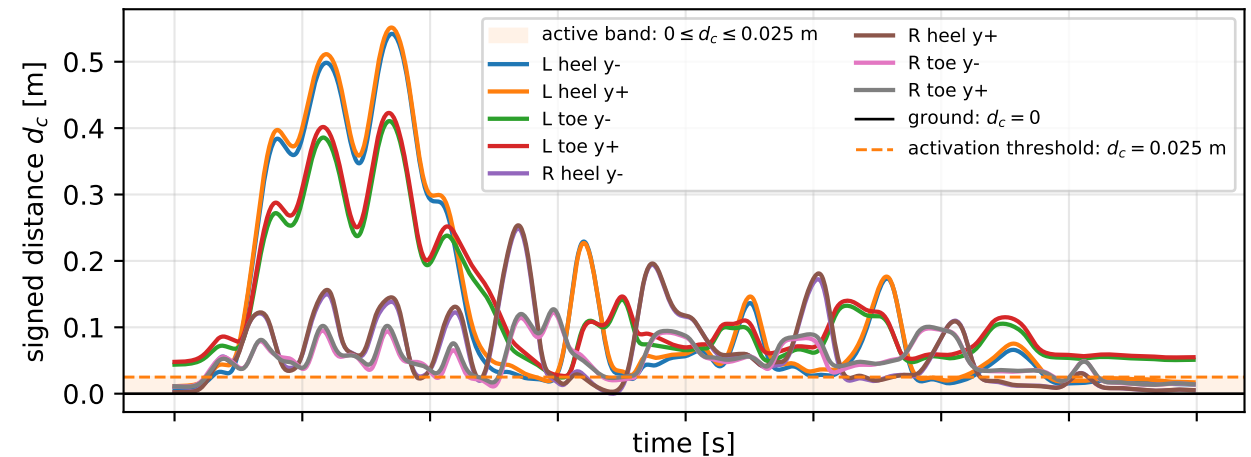


Fixed-contact force analysis: Transitions JOOF walk retargeted AMASS for Unitree G1

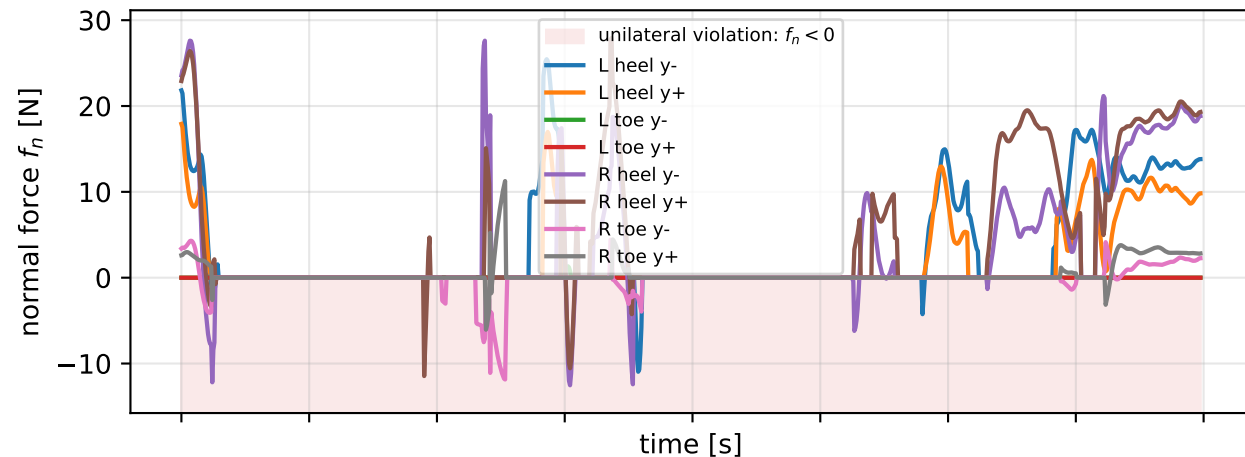
Detected fixed-contact set



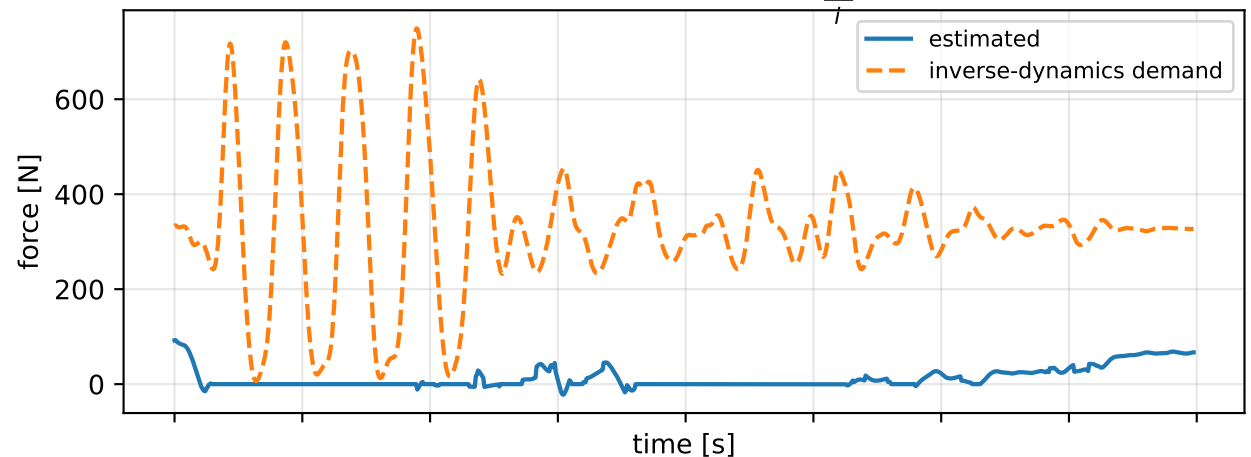
Contact heights and activation threshold



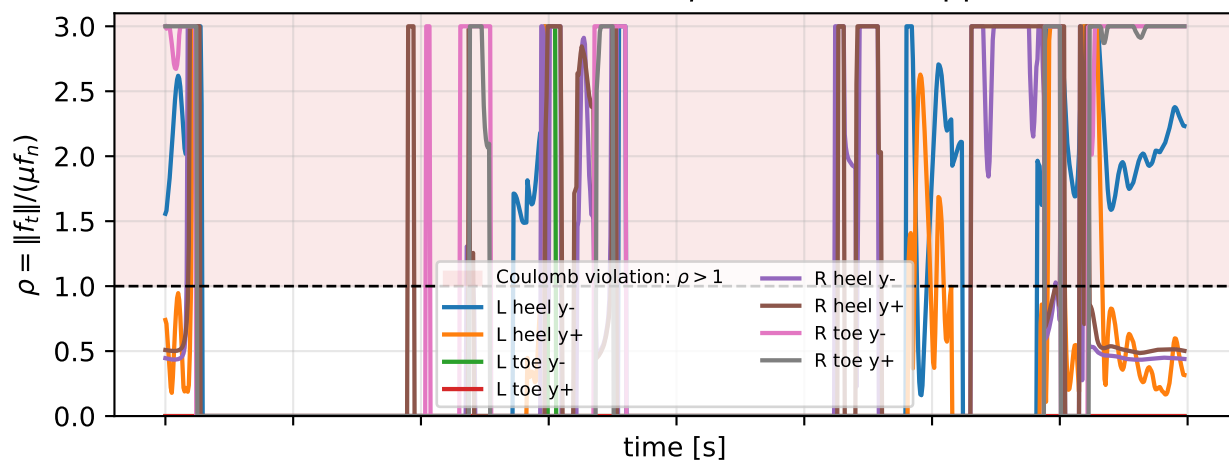
Per-contact normal forces



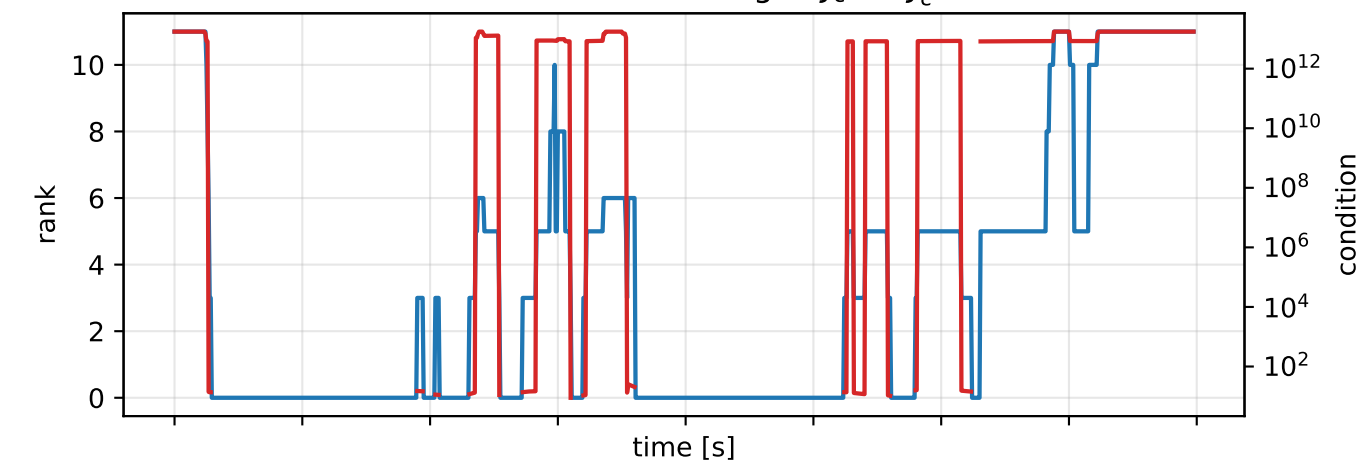
Total vertical support $\sum f_{n,i}$



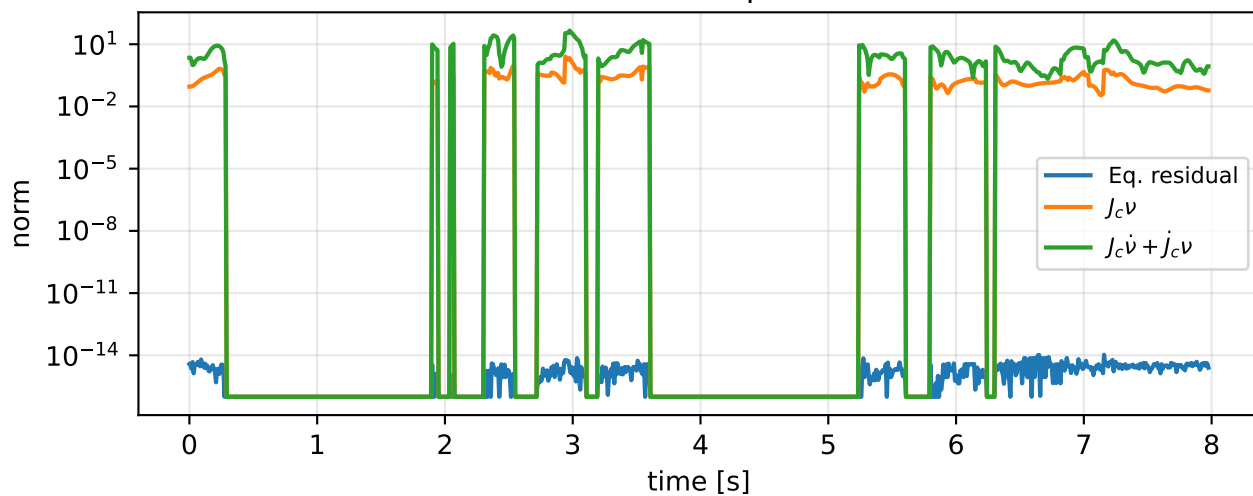
Friction ratio; shaded when $\rho > 1$ (values clipped at 3)



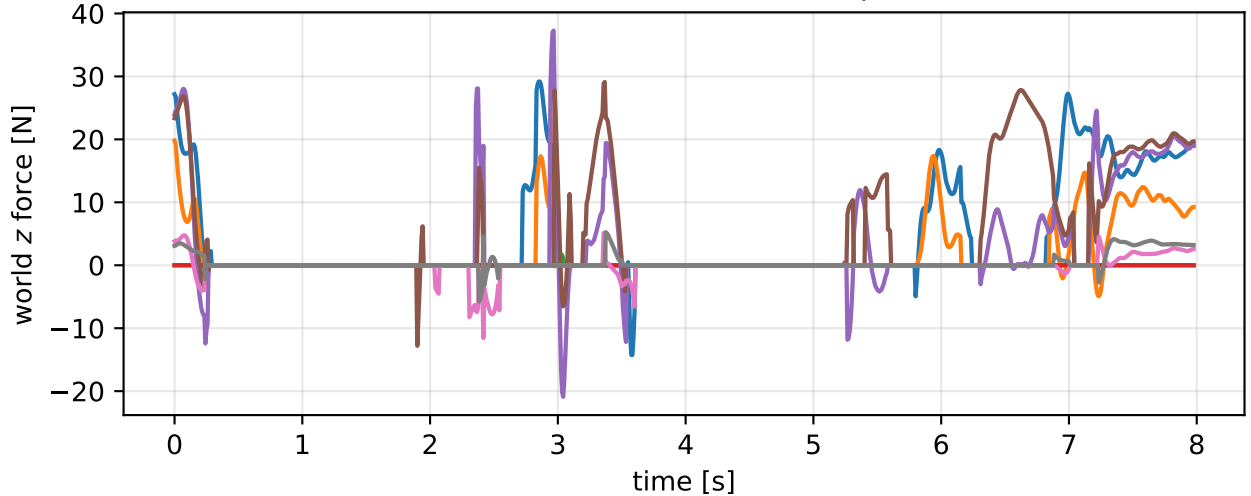
Rank and conditioning of $J_c M^{-1} J_c^T$



Fixed-contact assumption residuals



World-frame vertical force component f_z



Friction ratio: $\rho = \|f_t\|/(\mu f_n)$. The red shaded band marks $\rho > 1$, where the tangential force exceeds the Coulomb limit $\|f_t\| \leq \mu f_n$.